

Curriculum-Based Measurement for Written Expression (CBM-W):

The Supporting Research

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Case study: Annie

Annie is a 16-year old tenth grader who has yet to pass the statewide graduation test in writing. She has been placed in a standards English class in an effort to help him pass the high-stakes tests.

Her teacher, Mrs. Penn, decides to monitor his writing progress using CBM.

Mrs. Penn will use this progress monitoring data to decide whether her instruction is effective.

Inquiry Questions:

- What type of CBM task should Annie's teacher select?
- How should she administer and score Annie's CBM performance?
- How should she document his progress?
- How should she use the data to make instructional decisions?



CBM Research in Written Expression

- CBM research began in the 1970s-80s at the University of Minnesota Institute for Research on Learning Disabilities (IRLD).
  - IRLD researchers focused on elementary-level writers.
- Research has continued from the 1990s to the present.
  - This research has focused on both elementary & secondary levels, as well as on early writing.



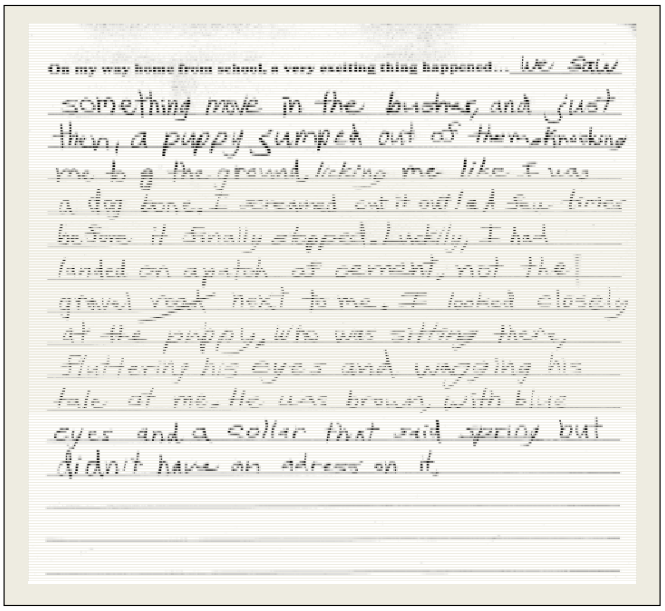
IRLD Dimensions of Assessing Writing:

Tasks	Duration	Scoring Procedures
Story starters/ narrative prompts	1 – 10 minutes	Words Written
Topic sentences/expository prompts		Words Spelled Correctly (WSC)
Picture/Photo prompts		Correct Word Sequences (CWS)

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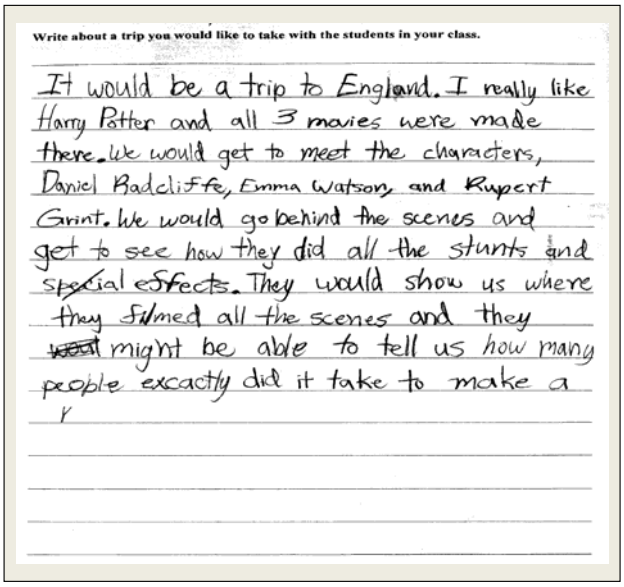
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Sample Narrative Prompt


On my way home from school, a very exciting thing happened... We saw something move in the bushes, and just then, a puppy jumped out of them. Knocking me to the ground, licking me like I was a dog bone. I screamed but it only led to more licks before it finally stopped. Luckily, I had landed on a patch of cement, not the gravel next to me. I looked closely at the puppy, who was sitting there, fluttering his eyes and wagging his tale at me. He was brown, with blue eyes and a collar that said spring but didn't have an adress on it.

(from RIPM CBM Writing Studies, Grades 3, 5, & 7)

Sample Expository Prompt



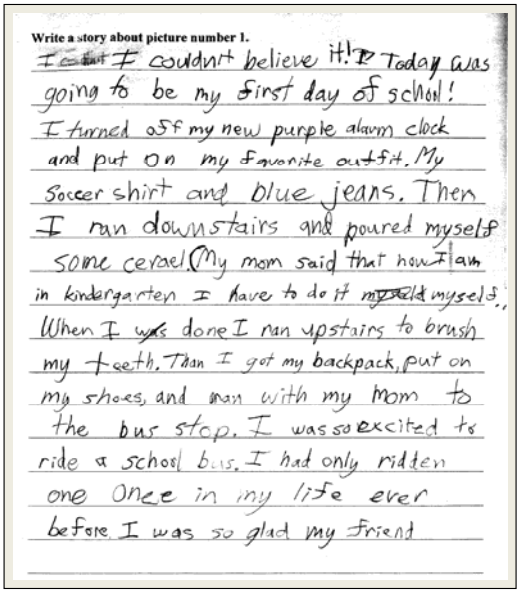
Write about a trip you would like to take with the students in your class.

It would be a trip to England. I really like Harry Potter and all 3 movies were made there. We would get to meet the characters, Daniel Radcliffe, Emma Watson, and Rupert Grint. We would go behind the scenes and get to see how they did all the stunts and special effects. They would show us where they filmed all the scenes and they might be able to tell us how many people excactly did it take to make a

(from RIPM CBM Writing Studies, Grades 3, 5, & 7)

Sample Picture Prompt





Write a story about picture number 1.

I couldn't believe it! Today was going to be my first day of school! I turned off my new purple alarm clock and put on my favorite outfit. My soccer shirt and blue jeans. Then I ran downstairs and poured myself some cereal. My mom said that now I am in kindergarten I have to do it myself. When I was done I ran upstairs to brush my teeth. Then I got my backpack, put on my shoes, and ran with my mom to the bus stop. I was so excited to ride a school bus. I had only ridden one Once in my life ever before I was so glad my friend

(from RIPM CBM Writing Studies, Grades 3, 5, & 7)

IRLD: Are CBM-W Measures Reliable?

Reliability

- Interrater
- Test-retest
- Alternate form

Most consistent findings:

- *Tasks:* Story starters
- *Duration:* 3-5 min
- *Scoring procedures:* WW, WSC, CWS



Deno et al., 1980; Fuchs et al., 1982; Marston & Deno, 1981; Marston et al., 1981; Shinn et al., 1982; Tindal, Marston, & Deno, 1983; Tindal, Germann, & Deno, 1983; Videen et al., 1982

IRLD: Are CBM-W Measures Valid?

Criterion Variables

- Standardized writing tasks (e.g., Test of Written Language)
- Developmental scoring systems
- Holistic ratings

Most consistent findings:

- *Tasks:* narrative, expository, picture
- *Durations:* 3-5 min
- *Scoring Procedures:* WW, WSC, CWS



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IRLD: Are CBM-W Measures Sensitive to Growth?

- Written expression measures:
  - Successfully discriminated among students of different age and skill levels.
  - Were sensitive to growth over time, both within and across grade levels .

Deno et al., 1980; Deno, Marston, Mirkin et al., 1982; Marston et al., 1983; Marston, Lowry, Deno, & Mirkin, 1981; Shinn et al., 1982

Extensions of IRLD Work

Fluency

- Based on ***amount*** of writing.
- WW, WSC, CWS

Accuracy

- Based on ***accuracy*** of writing.
- %WSC, %CWS

Tindal and Parker (1991); Parker, Tindal, & Hasbrouk (1991)

IRLD: Accuracy (percentage) measures

- Had stronger validity;
- Discriminated among low performers;
- Were good for screening;
- BUT, Difficult to monitor growth.

Tindal & Parker (1989); Tindal & Parker (1991); Parker, Tindal, & Hasbrouk (1991)

IRLD: Additional scoring procedures

- Other researchers have examined additional scoring procedures (e.g., punctuation, sentence length).
  - Weak to moderate reliability (*r*s < .65)
  - Weak validity coefficients (*r*s < .40)
  - WW, CWS most promising

Gansle, Noell, VanDerHeyden, Naquin, & Slider, (2002);
Gansle, Noell, VanDerHeyden, Slider, Hoffpauir, & Whitmarsh, et al. (2004)

IRLD: Gender differences and efficiency

- Girls write more than boys on fluency measures
- No differences on accuracy measures

- Scoring time (efficiency)
  - 3 min passages could be scored in < 5 min per child

Malecki & Jewell (2003); Jewell & Malecki (2005)

IRLD: More complex scoring procedures required at secondary level?

- A combination of characters per word, sentences, and mean length of correct word sequence strings.

- Difficult to score, graph, and interpret.

Espin, Scierka, Skare, & Halverson (1999)

IRLD: Correct Minus Incorrect Word Sequences

- Promising index for secondary-level students:
  - Moderate criterion validity (*r*s = .65 to .70)
  - More viable measure for detecting growth
  - Worked well for narrative and expository text
  - Reliability/validity increases with duration of sample (up to 7-10 min)

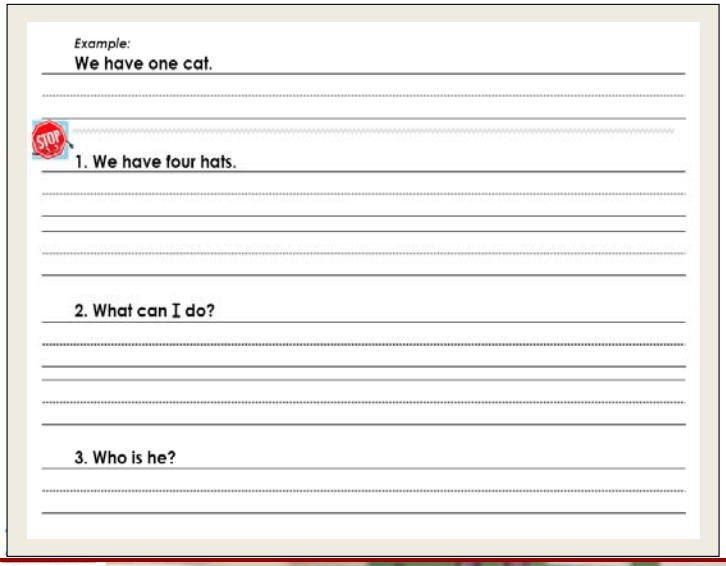
Espin, Shin, Deno, Skare, Robinson, & Benner (2000); Espin and Wallace et al. (2005)

Recent RIPM Findings: Grade 1

Measure	Reliable? (r > .70)	Criterion-Valid? (r > .50)	Avg. weekly growth from fall-spring analysis (from 5-min measures; p < .001)
Sentence Copying (3-5 min)	All scoring	All scoring	1.30 WW, 1.27 WSC, 1.47 CWS, 1.41 CIWS, 5.37 CLS
Story Prompt (5 min)	All scoring	WW, WSC, CWS, CLS	1.18 WW, 1.15 WSC, .67 CWS, 4.62 CLS
Picture-Word (3-5 min)	WW, WSC, CWS, CLS	WW, WSC, CWS, CLS	1.07 WW, 1.09 WSC, 1.11 CWS, 4.53 CLS
Photo Prompt (5 min)	WW, WSC, CWS, CLS	CWS, CLS	.58 CWS, 2.43 CLS
Note: N = 50. Criterion measures included district rubric, teacher ratings, TOWL-3			

McMaster, Du, and Petursdottir (2009)

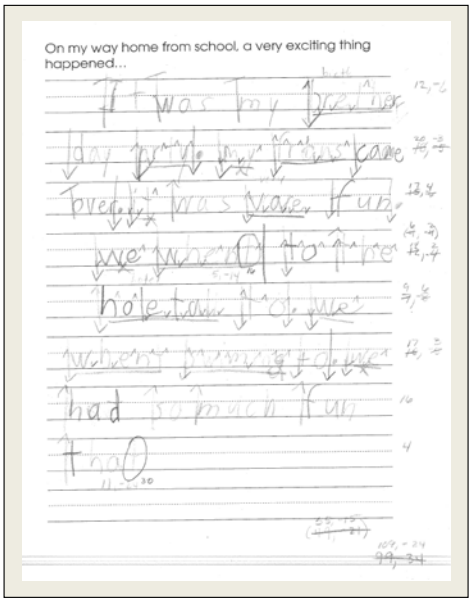
Sample Sentence Copy Prompt
(from RIPM CBM Writing Studies, Grade 1)



Sample Picture-Word Prompt
(from RIPM CBM Writing Studies, Grade 1)



Sample Story Prompt
(from RIPM CBM Writing Studies, Grade 1)



Translation:

It was my birthday party. My friends came over. It was very fun. We went to the hotel too. We went swimming too. We had so much fun that…

General Findings Within and Across Grades

Grade Level	Type of Prompt	Duration	Scoring Procedures
	Sentence Copying	3 to 5 min	WW, WSC, CWS
1-2	**Picture-Word**	**3 to 5 min**	**WW, WSC, CWS**
	Narrative	3 to 5 min	WW, WSC, CWS
3-4	Narrative	3 to 5 min	CWS, CIWS
5-6	Narrative	3 to 5 min	CWS, CIWS
	Expository	5 min	CWS, CIWS
7-9	Narrative	5 to 10 min	CWS, CIWS, %CWS
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WW = Words Written; WSC = Words Spelled Correctly, CWS = Correct Word Sequences; CIWS = Correct minus Incorrect Word Sequences

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Summary: What does research tell us to date?

What we know:

- CBM-W generally produces reliable and criterion-valid scores
- CBM-W appears to be sensitive to growth over time

What we still need to learn:

- We are still examining characteristics of slopes produced from progress-monitoring data
- We are still examining effects of teachers' use of the data to make instructional decisions

For a literature review, see McMaster & Espin (2007)

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Three Cautions

- Selection of measures will depend on the characteristics of the students in the class.
- To measure performance and progress in writing, teachers must think carefully about how they or their district define "good writing."
- Because little work has been done on the use of the measures as progress measures, some information, for example typical growth rates, may need to be determined locally.

TIP

Continued research is being conducted on CBM-W. It is best to keep abreast of the continued development of CBM-W measures.



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